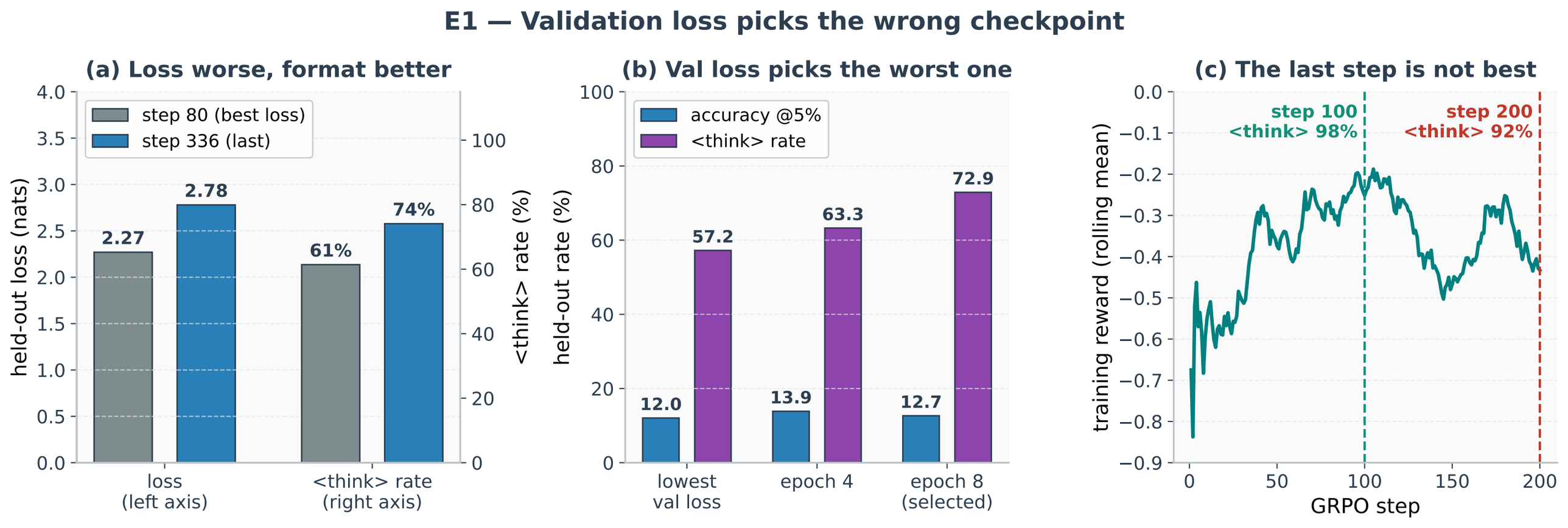
E1 — Validation loss picks the wrong checkpoint
(a) Loss worse, format better
step 80 (best loss)
step 336 (last)
held-out loss (nats)
<think> rate (%)
2.27
2.78
61%
74%
loss
(left axis)
<think> rate
(right axis)
(b) Val loss picks the worst one
accuracy @5%
<think> rate
held-out rate (%)
12.0
57.2
13.9
63.3
12.7
72.9
lowest
val loss
epoch 4
epoch 8
(selected)
(c) The last step is not best
step 100
<think> 98%
step 200
<think> 92%
training reward (rolling mean)
GRPO step